RACE
Regular Analysis Through Continuous Exercise

Errors in Measurements and Instruments : RACE-01

1. Match List-I with List-II.

List-I (Number)	List-II (Significant figure)
(a) 1001	(I) 3
(b) 010.1	(II) 4
(c) 100.100	(III) 5
(d) 0.0010010	(IV) 6

Choose the correct answer from the options given below:

(A) (a)-(III), (b)-(IV), (c)-(II), (d)-(I) (B) (a)-(IV), (b)-(III), (c)-(I), (d)-(II)

(C) (a)-(II), (b)-(I), (c)-(IV), (d)-(III) (D) (a)-(I), (b)-(II), (c)-(III), (d)-(IV)

PER125

2. The resistance $R=\frac{V}{I}$ where $V=(200\pm5)V$ and $I=(20\pm0.2)A$, the percentage error in the measurement of R is :

(A) 3.5% (B) 7% (C) 3% (D) 5.5%

PER126

3. In a simple pendulum experiment for determination of acceleration due to gravity (g), time taken for 20 oscillations is measured by using a watch of 1 second least count. The mean value of time taken comes out to be 30s. The length of pendulum is measured by using a meter scale of least count 1 mm and the value obtained is 55.0 cm. The percentage error in the determination of g is close to :-

(A) 0.7% (B) 0.2% (C) 3.5% (D) 6.8%

PER127

4. The density of a solid metal sphere is determined by measuring its mass and its diameter. The maximum error in the density of the sphere is $\left(\frac{x}{100}\right)\%$. If the relative errors in measuring the mass and the diameter are 6.0% and 1.5% respectively, the value of x is

PER128

5. The period of oscillation of a simple pendulum is $T=2\pi\sqrt{\frac{L}{g}}$. Measured value of 'L' is 1.0 m from meter scale having a minimum division of 1 mm and time of one complete oscillation is 1.95s measured from stopwatch of 0.01s resolution. The percentage error in the determination of 'g' will be :

(A) 1.13% (B) 1.03% (C) 1.33% (D) 1.30%

PER129

6. In order to determine the Young's Modulus of a wire of radius 0.2 cm (measured using a scale of least count = 0.001 cm) and length 1m (measured using a scale of least count = 1 mm), a weight of mass 1 kg (measured using a scale of least count = 1g) was hanged to get the elongation of 0.5 cm (measured using a scale of least count 0.001 cm). What will be the fractional error in the value of Young's Modulus determined by this experiment?

(A) 0.14% (B) 0.9% (C) 9% (D) 1.4%

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7. Three students S_1, S_2 and S_3 perform an experiment for determining the acceleration due to gravity (g) using a simple pendulum. They use different lengths of pendulum and record time for different number of oscillations. The observations are as shown in the table.

Student No.	Length of pendulum (*cm*)	No. of oscillations (*n*)	Total time for *n* oscillations	Time period (*s*)
1.	64.0	8	128.0	16.0
2.	64.0	4	64.0	16.0
3.	20.0	4	36.0	9.0

(Least count of length = 0.1 *m* least count for time = 0.1 *s*)

If E_1, E_2 and E_3 are the percentage errors in 'g' for students 1, 2 and 3 respectively, then the minimum percentage error is obtained by student number _____.

PER131

8. Two resistors $R_1 = (4 \pm 0.8)\ \Omega$ and $R_2 = (4 \pm 0.4)\Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be :

(A) $(4 \pm 0.4)\ \Omega$ (B) $(2 \pm 0.4)\ \Omega$ (C) $(2 \pm 0.3)\ \Omega$ (D) $(4 \pm 0.3)\ \Omega$

PER132

9. A 2 meter long scale with least count of 0.2 cm is used to measure the locations of objects on an optical bench. While measuring the focal length of a convex lens, the object pin and the convex lens are placed at 80 cm mark and *lm* mark, respectively. The image of the object pin on the other side of lens coincides with image pin that is kept at 180 cm mark. The % error in the estimation of focal length is:

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(A) 1.02 (B) 0.85 (C) 1.70 (D) 0.51

PER133

10. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Assertion A : A spherical body of radius (5 ± 0.1) mm having a particular density is falling through a liquid of constant density. The percentage error in the calculation of its terminal velocity is 4%.

Reason R : The terminal velocity of the spherical body falling through the liquid is inversely proportional to its radius.

In the light of the above statements, choose the ***correct*** answer from the options given below:

(A) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A.**

(B) Both **A** and **R** are true and **R** is the correct explanation of **A.**

(C) **A** is false but **R** is true.

(D) **A** is true but **R** is false.

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	A	D	1050	A	D	1	C	C	D

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Regular Analysis Through Continuous Exercise

Errors in Measurements and Instruments : RACE-02

1. Given below are two **statements**: one is labelled as Assertion (A) and the other is labelled as **Reason (R).**

Assertion (A) : In Vernier calliper if positive zero error exists, then while taking measurements, the reading taken will be more than the actual reading.

Reason (R) : The zero error in Vernier Calliper might have happened due to manufacturing defect or due to rough handling.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(B) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

PER135

2. The pitch and the number of divisions, on the circular scale, for a given screw gauge are 0.5 *mm* and 100 respectively. When the screw gauge is fully tightened without any object, the zero of its circular scale lies 3 divisions below the mean line.

The readings of the main scale and the circular scale, for a thin sheet, are 5.5 *mm* and 48 respectively, the thickness of this sheet is:

(A) 5.755 *m* (B) 5.725 *mm* (C) 5.740 *m* (D) 5.950 *mm*

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3. The least count of the main scale of a vernier callipers is 0.1 *mm*. Its vernier scale is divided into 10 divisions and coincide with 9 divisions of the main scale. When jaws are touching each other, the 7th division of vernier scale coincides with a division of main scale and the zero of vernier scale is lying right side of the zero of main scale. When this vernier is used to measure length of a cylinder the zero of the vernier scale between 3.1 *cm* and 3.2 *cm* and 4th VSD coincides with a main scale division. The length of the cylinder is : (VSD is vernier scale division)

(A) 3.21 *cm* (B) 2.99 *cm* (C) 3.2 *cm* (D) 3.07 *cm*

PER137

4. The pitch of the screw gauge is 1 *mm* and there are 100 divisions on the circular scale. When nothing is put in between the jaws, the zero of the circular scale lies 8 divisions below the reference line. When a wire is placed between the jaws, the first linear scale division is clearly visible while 72nd division on circular scale coincides with the reference line. The radius of the wire is:

(A) 1.64 *mm* (B) 0.82 *mm* (C) 1.80 *mm* (D) 0.90 *mm*

PER138

5. One main scale division of a vernier callipers is 'a' *cm* and n^{th} division of the vernier scale coincide with $(n-1)^{th}$ division of the main scale. The least count of the callipers in *mm* is :

(A) $\frac{10na}{n-1}$ (B) $\frac{10a}{(n-1)}$ (C) $\left(\frac{n-1}{10n}\right)a$ (D) $\frac{10a}{n}$

PER139

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6. The vernier scale used for measurement has a positive zero error of 0.2 *mm*. If while taking a measurement it was noted that '0' on the vernier scale lies between 8.5 *cm* and 8.6 *cm*, vernier coincidence is 6, then the correct value of measurement is____ *cm*. (Least count = 0.01 *cm*)

(A) 8.36 *cm* (B) 8.54 *cm* (C) 8.58 *cm* (D) 8.56 *cm*

PER140

7. In a Screw Gauge, fifth division of the circular scale coincides with the reference line when the ratchet is closed. There are 50 divisions on the circular scale, and the main scale moves by 0.5 *mm* on a complete rotation. For a particular observation the reading on the main scale is 5 *mm* and the 20th division of the circular scale coincides with reference line. Calculate the true reading.

(A) 5.00 *mm* (B) 5.25 *mm* (C) 5.15 *mm* (D) 5.20 *mm*

PER141

8. The diameter of a spherical bob is measured using a vernier callipers. 9 divisions of the main scale, in the vernier callipers, are equal to 10 divisions of vernier scale. One main scale division is 1 *mm*. The main scale reading is 10 *mm* and 8th division of vernier scale was found to coincide exactly with one of the main scale division. If the given vernier callipers has positive zero error of 0.04 *cm*, then the radius of the bob is _____ $\times 10^{-2}$ *cm*.

PER142

9. A student in the laboratory measures thickness of a wire using screw gauge. The readings are 1.22 *mm*, 1.23 *mm*, 1.19 *mm* and 1.20 *mm*. The percentage error is $\frac{x}{121}$ %. The value of *x* is _____.

PER143

10. In an experiment with Vernier callipers of least count 0.1 *mm*, when two jaws are joined together the zero of Vernier scale lies right to the zero of the main scale and 6th division of Vernier scale coincides with the main scale division. While measuring the diameter of a spherical bob, the zero of vernier scale lies in between 3.2 *cm* and 3.3 *cm* marks, and 4th division of vernier scale coincides with the main scale division. The diameter of bob is measured as:

(A) 3.18 *cm* (B) 3.25 *cm* (C) 3.26 *cm* (D) 3.22 *cm*

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	B	D	B	D	B	C	52	150	A

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